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ALL DAY DEVOPS

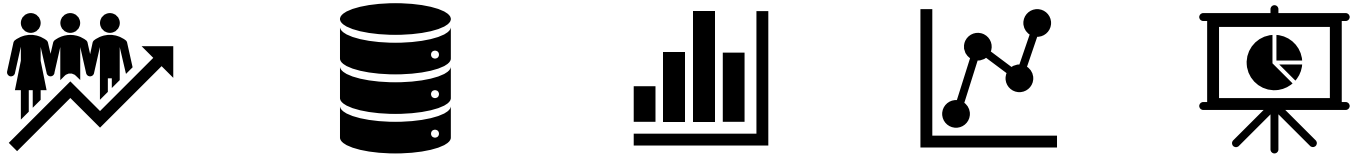
NOVEMBER 12, 2020

Our DevOps Journey is Incomplete Without Data

Our Data, Architecture & Data Management Practices Need Love & Attention

BMK LAKSHMINARAYANAN(@LBMKRISHNA)
Value Stream Architect @BNZ

Data is the lifeblood of the system of the business



Flow

Problem Context	Context, Statistics, Impact
Challenges	Ownership, Silo, Locking, Standards
Solution Approach	Education, Conventions, Architecture, Refactor
Recommendations	Useful Resources
Remaining Challenges	Call for Action

ABOUT ME

BMK Lakshminarayanan
Solutions Architect | DevOps Advocate

MISSION

Make your life better,
Make others life better,
Make the world better

Community

DEVOPS DAYS
CLOUD NATIVE SUMMIT
CLOUD NATIVE COMPUTING FOUNDATION AMBASSADOR
AMBASSADOR DevOps INSTITUTE Human of DevOps
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Understanding Our Problem

Problem Context

Continuous Delivery, DevOps & Data

Problem Context

- Manual database deployments continue to be a Software delivery bottleneck
- Speeding up database releases are most difficult
- Neglecting the data in our DevOps, Continuous Delivery processes has significant “Business Impact”
- Data issues continue to slow us down, trouble our efforts to continuously integrate, test & deploy

Statistics

46% Extremely or very difficult to speed database releases

23% High risk of deployment failure & extra downtime due to database changes

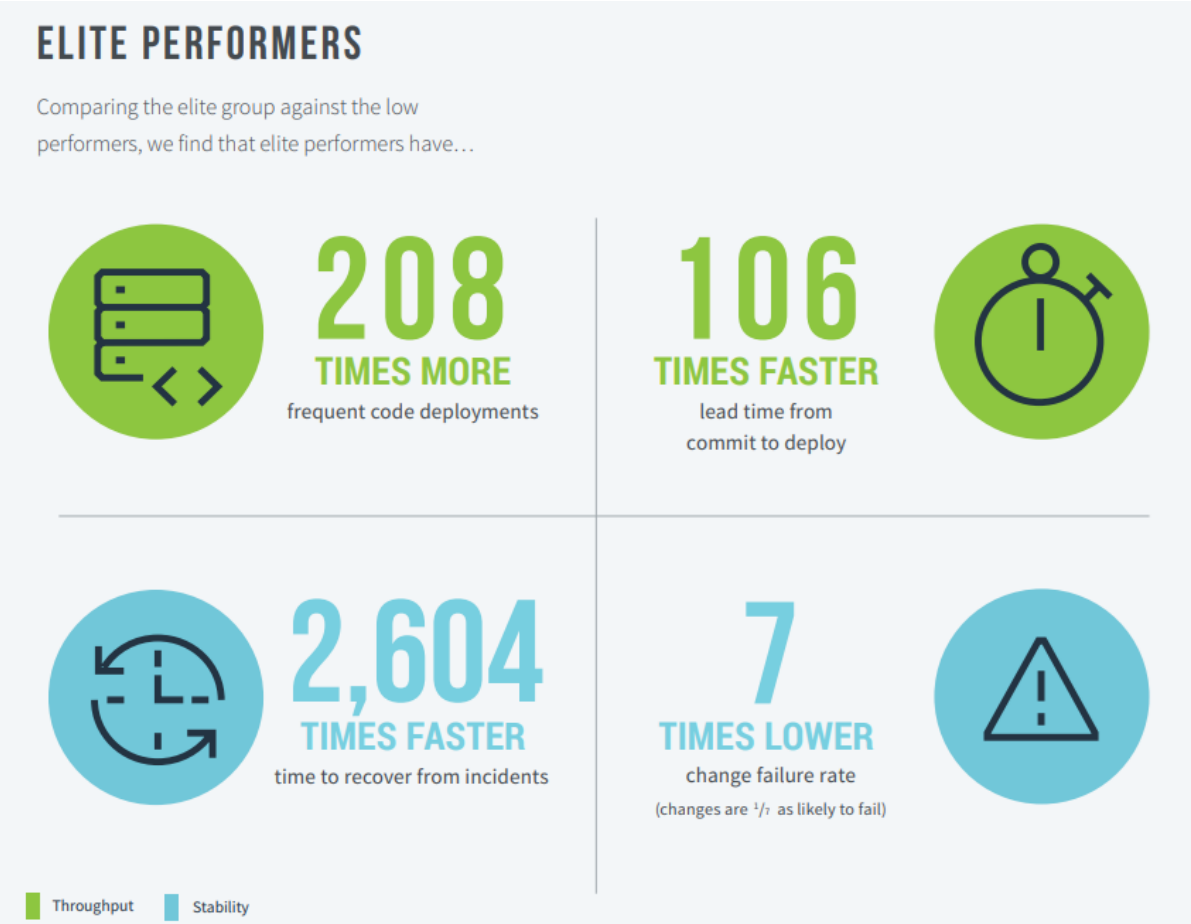
20% We are slow due to traditional siloed database development practices

57% Application changes require corresponding database changes

Source :https://www.devopsdigest.com/the-state-of-database-deployments-in-application-delivery-2019

Yesterday's DBA

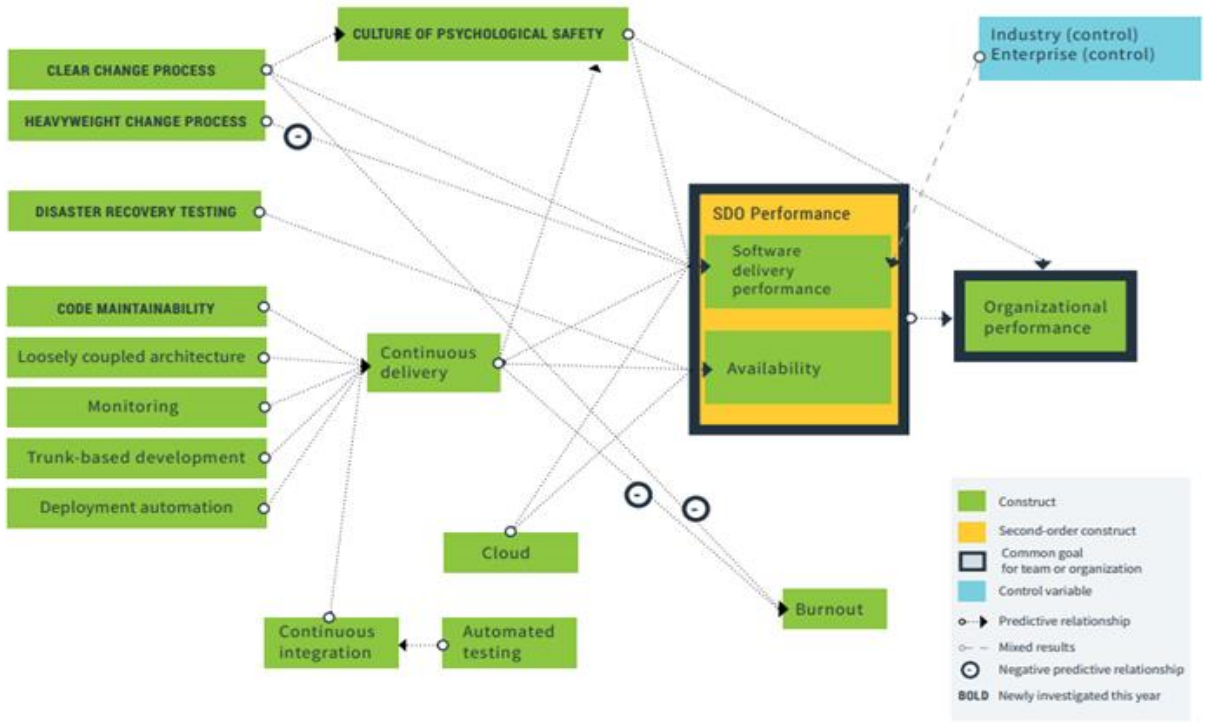
State of DevOps Report 2019



https://services.google.com/fh/files/misc/state-of-devops-2019.pdf

How do we compare?

State of DevOps Report 2019



IT Performance => Organization Performance

Accelerate: State of DevOps 2019 | How Do We Improve SDO & Organizational Performance?

https://services.google.com/fh/files/misc/state-of-devops-2019.pdf

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Understanding Our Challenges

Challenges

Continuous Delivery, DevOps & Data

Enterprise Data Challenges

- How do we think about data? Data lock-in
- Decade old standards & conventions
- Inherited data management practices
- Centralize data team to design, model, deploy & operate

Who is Responsible?

Our system complexities are because of our thinking, thinking particularly about the "Data". We tend to compromise a lot as we could not get the "Data" in the right way, at the right time, in the proper format, to the right place

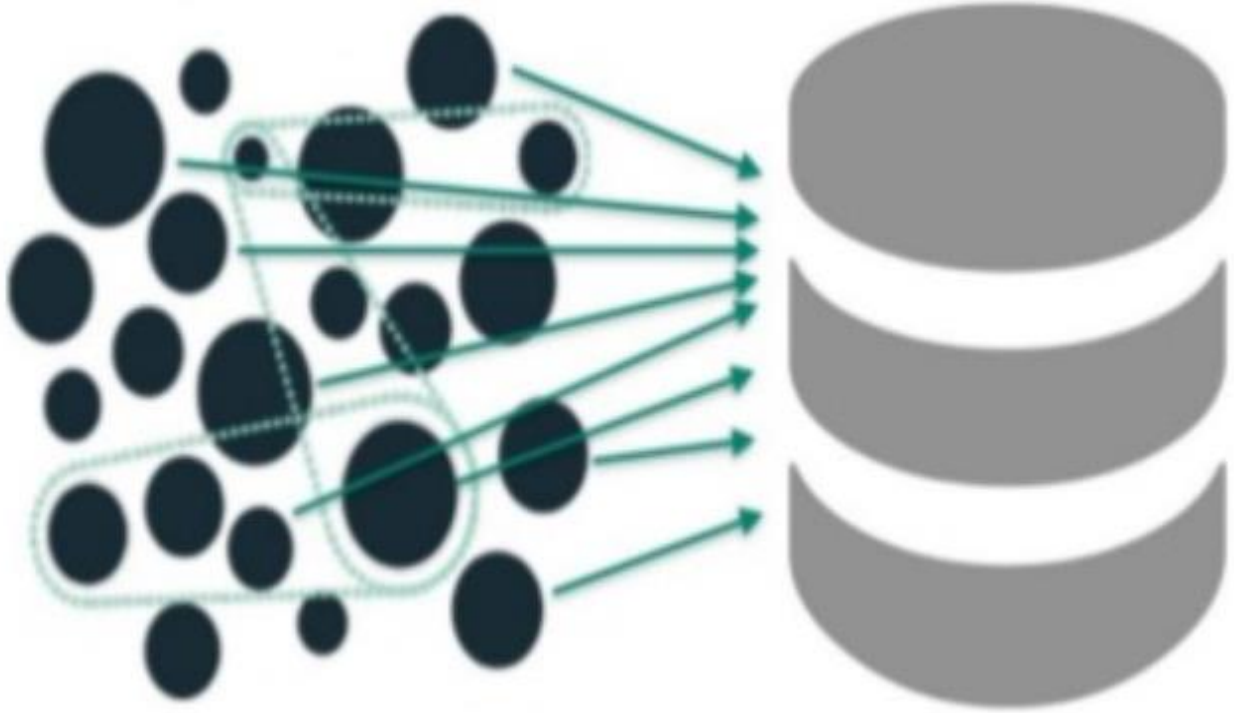
Systems of Engagement

Systems of Intelligence

Systems of Record

Architecture Mismatch

Microservices Era & Big Ball of "Data" mud



Mom asked me to share Database

Small Steps, But in Right Direction

Solution Approach

Education, Conventions, Architecture, Shift Left

Solution Approach

- Data Management Team
  Decentralize
- Conventions & Standards
- Educate, Upskill, Tools,
  Processes
- Unlock the Data
  Adopting Modern Practices

Solution Approach

12%

Poor - DBAs are completely siloed, unaware of what's happening in development, and deploy changes to production in isolation

23%

Great - all work together effectively as one team, using standardized processes and tools

25%

OK - DBAs aren't really involved and only work with development when it's time for production deployment

40%

Good - teams collaborate, with DBAs involved on an ad hoc basis when needed

http://assets.red-gate.com/solutions/database-devops/state-of-database-devops-2020.pdf?v=2

Help Our Data Folks - Practices

- Make the data first-class citizen
- Source control - Database schema
- Simplify processes
- Review changes early
- CI/CD pipeline

Help Our Data Folks - Futuristic

- “T” & “Comb” shaped
- Data Engineers
- Educator & Mentors
- Platform Builders

Help Our Dev & Ops Folks - Educate

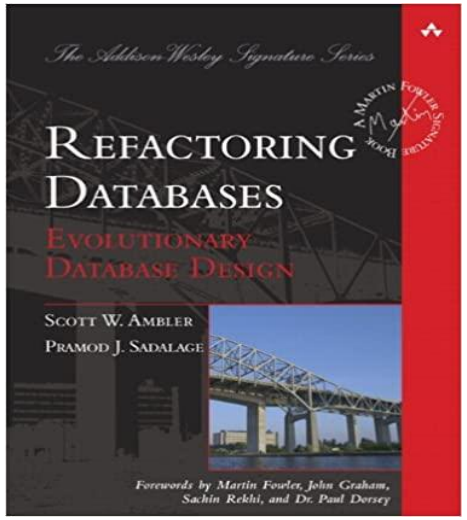
- Data Governance, Lineage
- Compliance (PCI DSS, Data Privacy)
- Regulations (GDPR, CCPA)
- Data Security (PII)
- Standards & Conventions (Naming, model, dictionary, catalogue)
- Enable Self-Service (Database request, Schema change request)

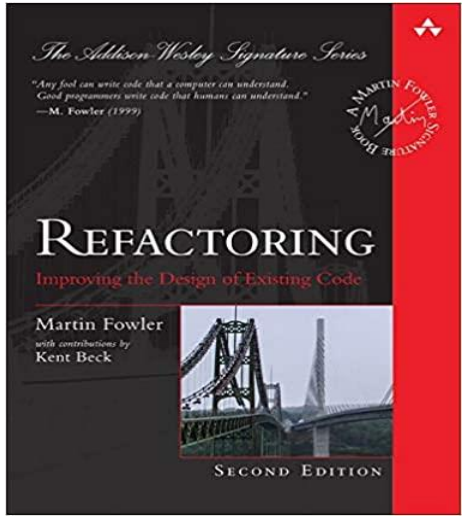
Useful Resources

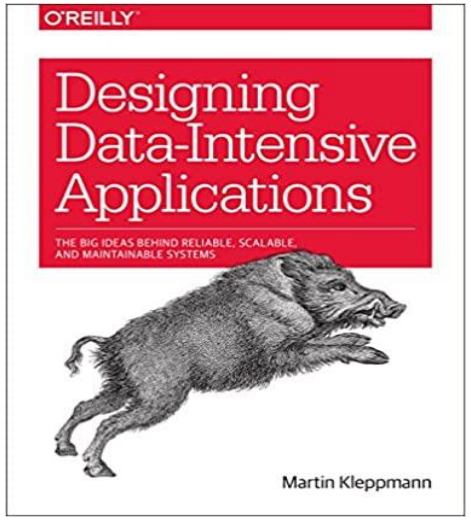
Recommendations

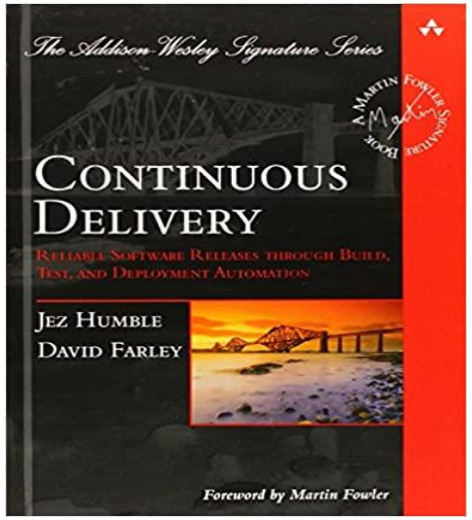
Upskill, Learn, Spread, Share

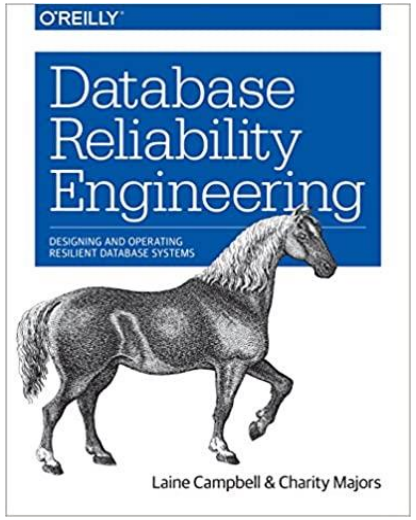
Recommendations











We need to level up our game
Data, Data Architecture, Data Management & Associated Practices need our love & attention

Recommended Approach

- Understanding current state (Tools, processes, roles, flow)
- Making work visible (current workloads, workloads in pipeline)
- Value Stream Management (New feature, Tech debt, Incident)
- Upskilling (DevOps, CI/CD Practices, Automation)
- Investment (Education, Time, Tools, Self-service)
- Role play (Data Engineer/Developer for a day)

Collectively We Need To Come Together, Work Together, Collaborate

Remaining Challenges

Call for Action

Help – Let Us Come Together

- Automated Governance (Including Data)
- Policies via automation and in pipeline
- Compliance as code
- Everyone is custodian of the Data

My dream
&
My vision

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